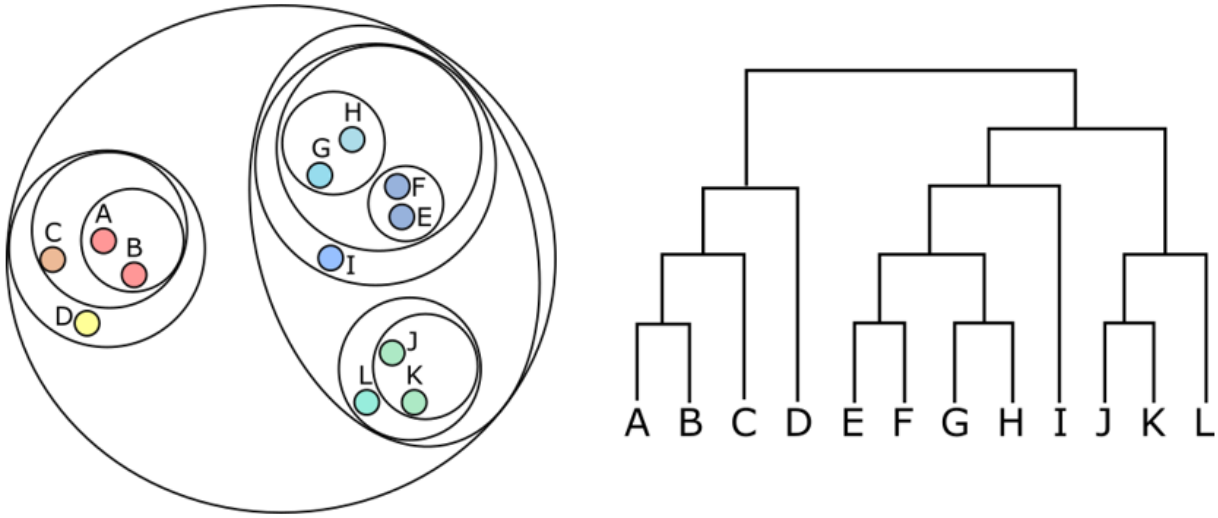


Figure 5.14 An example of hierarchical clustering. On the left, the dataset (a collection of 2D points) is shown with progressive grouping shown as ellipses. From the figure, it can be inferred that, for instance, A and B are grouped together before C is added to the two of them to form a larger cluster. Hence, the relationship between A and B is inferred to be stronger than the one between A and C or A and B. On the right, the same process is shown using a *dendrogram*[16].
Note that both figures could be the result either of agglomerative or divisive clustering: the former produces the dendrogram starting from the bottom, the latter starting from the top.

The description of the algorithm should ring a bell: at each step we need to find two points belonging to two different clusters. You can easily imagine what's the best data structure to compute and find that information. In chapter 13, we'll see a practical application of disjoint set as part of a distributed clustering algorithm.